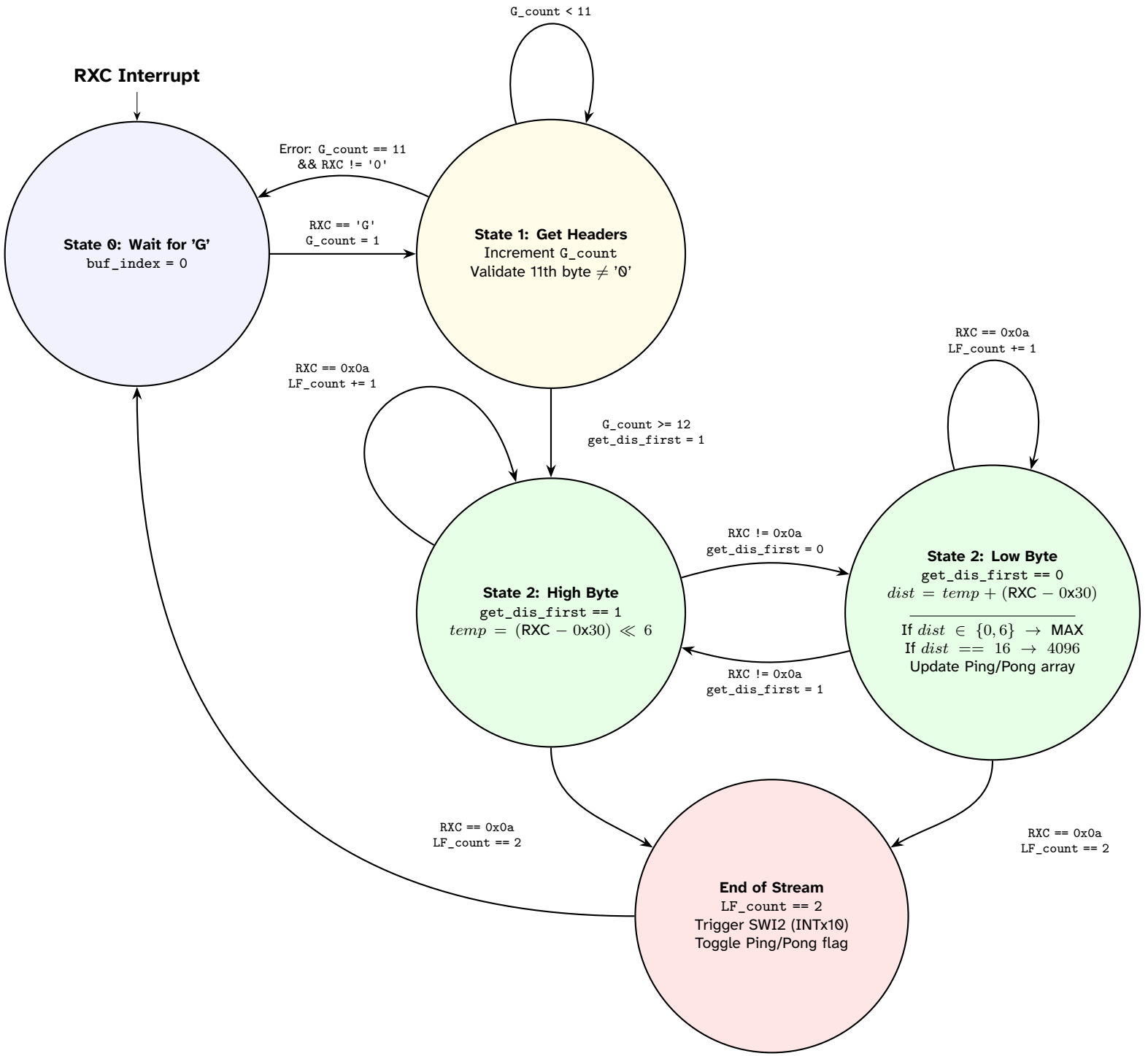
RXC Interrupt
State 0: Wait for 'G'
buf_index = 0
RXC == 'G'
G_count = 1
State 1: Get Headers
Increment G_count
Validate 11th byte ≠ '0'
G_count < 11
Error: G_count == 11
&& RXC != '0'
G_count >= 12
get_dis_first = 1
State 2: High Byte
get_dis_first == 1
temp = (RXC − 0x30) ≪ 6
RXC == 0x0a
LF_count += 1
RXC != 0x0a
get_dis_first = 0
RXC != 0x0a
get_dis_first = 1
State 2: Low Byte
get_dis_first == 0
dist = temp + (RXC − 0x30)
If dist ∈ {0, 6} → MAX
If dist == 16 → 4096
Update Ping/Pong array
RXC == 0x0a
LF_count += 1
RXC == 0x0a
LF_count == 2
RXC == 0x0a
LF_count == 2
End of Stream
LF_count == 2
Trigger SWI2 (INTx10)
Toggle Ping/Pong flag